

Final Examination
Introduction to Massive Data Analysis

Dec. 9, 2020

1. (24%) Yes/no questions. You don't have to justify the answers.

- ☒ (a) A word has high TF.IDF score in a document if it appears in relatively ~~few~~ documents, but appears in this one, and when it appears in a document it tends to appear less times. $\frac{f}{\max f}$ $\frac{1}{n}$
- ☒ (b) Bloom filters guarantee no false positives.
- ☐ (c) A set of basis functions to start LSH for Euclidean distance can be obtained by choosing random lines and projecting points onto those lines.
- ☐ (d) A good estimate for the second moment (surprise number) is obtained by choosing several fixed positions in the stream, taking twice the number of times this element appears in the stream from that position onward, subtracting 1, and multiplying by the length of the stream.
- ☒ (e) A family F of functions is said to be (d_1, d_2, p_1, p_2) -sensitive if for every f in F . If $d(x, y) \leq d_1$, then the probability that $f(x) = f(y)$ is ~~at most~~ p_1 . NO
- ☐ (f) For each AND-OR S -curve $1 - (1 - s^r)^b$, there is a threshold t , for which $1 - (1 - t^r)^b = t$. Above t , high probabilities are decreased; below t , low probabilities are increased.
- ☐ (g) In UCB, confidence interval grows with the total number of actions t we have taken, but shrinks with the number of times m_a we have tried arm a .
- ☒ (h) To identify spam farms, we can compute both the conventional PageRank and the TrustRank for all pages. Those pages that have much higher TrustRank than PageRank are likely to be part of a spam farm.
- ☐ (i) PageRank was used to simulate where Web surfers, starting at a random page, would tend to congregate if they followed randomly chosen outlinks from the page at which they were currently located, and this process were allowed to iterate many times.
- ☒ (j) Instead of making passes through all the data, we may choose a random sample of the baskets, small enough that it is possible to store both the sample and the needed counts of itemsets in main memory. We can then find the frequent itemsets for the sample, and hope that it is a good representation of the data as whole. While this method uses at most one pass through the whole dataset, it is subject to false positives only.
- ☒ (k) Many on-line algorithms are greedy, in the sense that they select their action at every step by minimizing some objective function.
- ☒ (l) One way of predicting the blank values in a utility matrix is to find two long, thin matrices U and V , whose product is an approximation to the given utility matrix. The intuitive reason this method makes sense is that often there are a

relatively small number of issues that determine whether or not a user likes an item.

2. (4%) There are many different approaches to modeling data. Most approaches (mentioned in this course) to modeling can be described as either a) Summarizing the data succinctly and approximately, or b) Extracting the most prominent features of the data and ignoring the rest. Which of the following four forms: frequent itemsets, similar items, web mining, clustering, are recognized as the form of Summarizing?
3. (4%) Show that e (natural logarithm) is the limit of $(1+1/x)^x$ as x goes to infinity.
4. (4%) Consider the following matrix:

	C1	C2	C3	C4
R1	1	0	0	1
R2	1	0	1	0
R3	0	1	1	0
R4	0	1	1	0
R5	0	1	1	0
R6	1	0	0	1
R7	1	0	0	1

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x &= y = \left(1 + \frac{1}{x}\right)^x \\ \ln y &= x \ln \left(1 + \frac{1}{x}\right) \\ &= \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}} \\ &= \frac{\frac{1}{1+\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} \end{aligned}$$

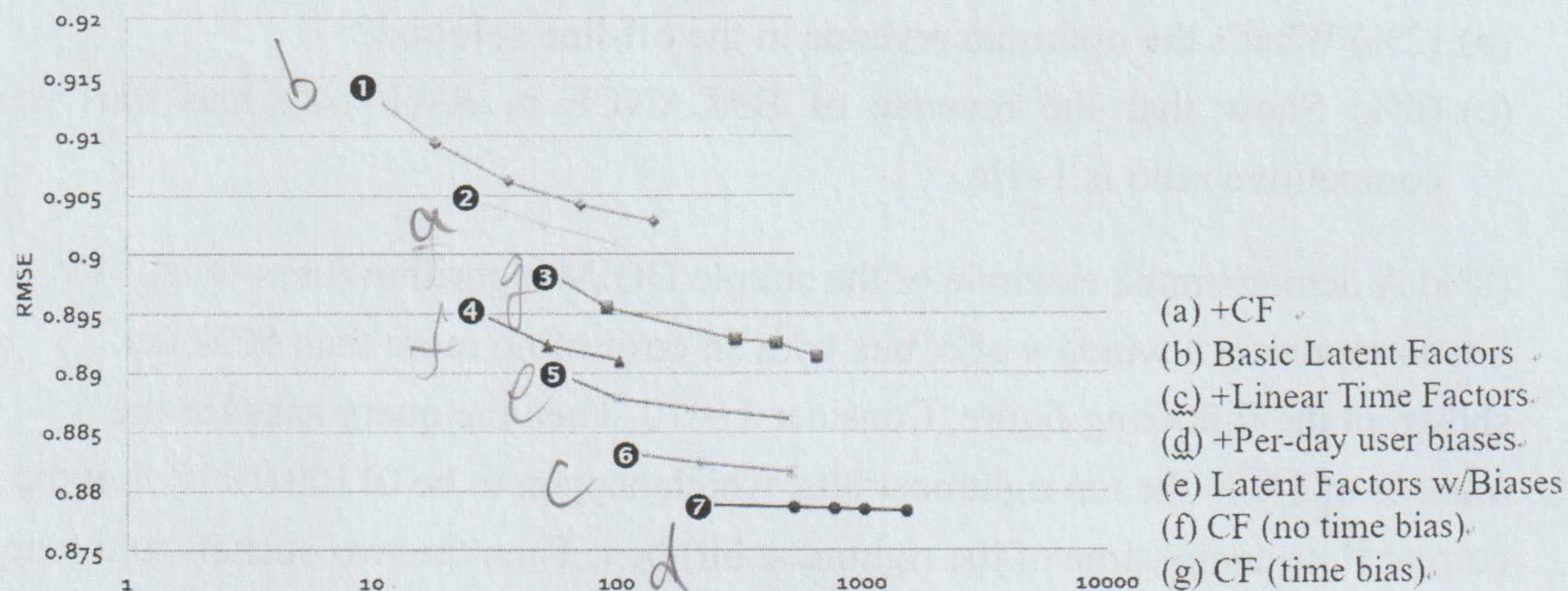
Perform a minhashing of the data, with the order of rows: $\pi = (R3, R4, R7, R2, R6, R1, R5)$. Compute the minhash value $h_\pi(C_i)$ for all C_i 's, and compute the similarities (col/col and sig/sig) between each pair of C_i and C_j .

5. (6%) The *reservoir sampling* algorithm can maintain a uniformly chosen sample of positions in the stream, that is, after n elements the sample contains each element seen so far with probability s/n , where s is the size of random samples. Prove this using proof by induction.
6. (4%) An association rule can have several items on the right as well as on the left. The confidence of an association rule $S \rightarrow T$ is the fraction of baskets containing all the items in set S that also contain all the items in set T . Suppose we have 7 items numbered as 1, 2, ..., 7. Which of the following association rules has a confidence that is certain to be at least as great as the confidence of $1,2 \rightarrow 3,4,5,6,7$ and no greater than the confidence of $1,2,3,4 \rightarrow 5$? Circle all and only those that do.

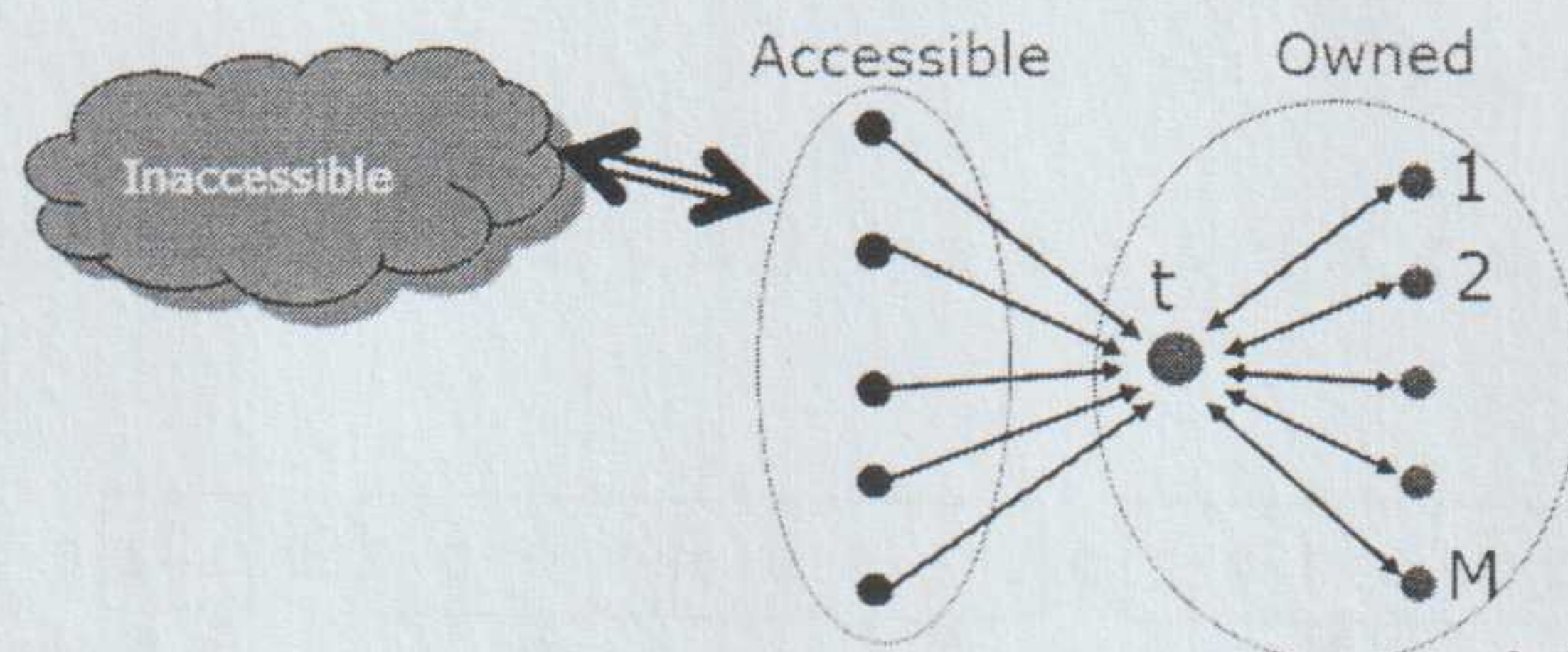
(a) $1,2,4 \rightarrow 3,6,7$ (b) $1,2,3 \rightarrow 4,5,7$ (c) $2,3,4 \rightarrow 1,5,7$ (d) $1,2,4 \rightarrow 3,5,6$.

7. (4%) The Three-stage Algorithm improves upon PCY by using two successive hash tables to reduce further the number of candidate pairs. The tradeoff is that Three-stage takes more than two passes to find the frequent pairs. In Pass 3, we count pairs $\{i, j\}$ if and only if: (i) both i, j are frequent, (ii) $\{i, j\}$ hashes to freq. bucket in B1, and (iii) $\{i, j\}$ hashes to freq. bucket in B2. Notice that all three conditions must be checked on the counting pass. We cannot eliminate the (ii) requirement in Pass 3, explain why.

8. (7%) Match (連連看) the items in the following figure.



9. (8%) One of the most common and effective organizations for a link farm is shown below:



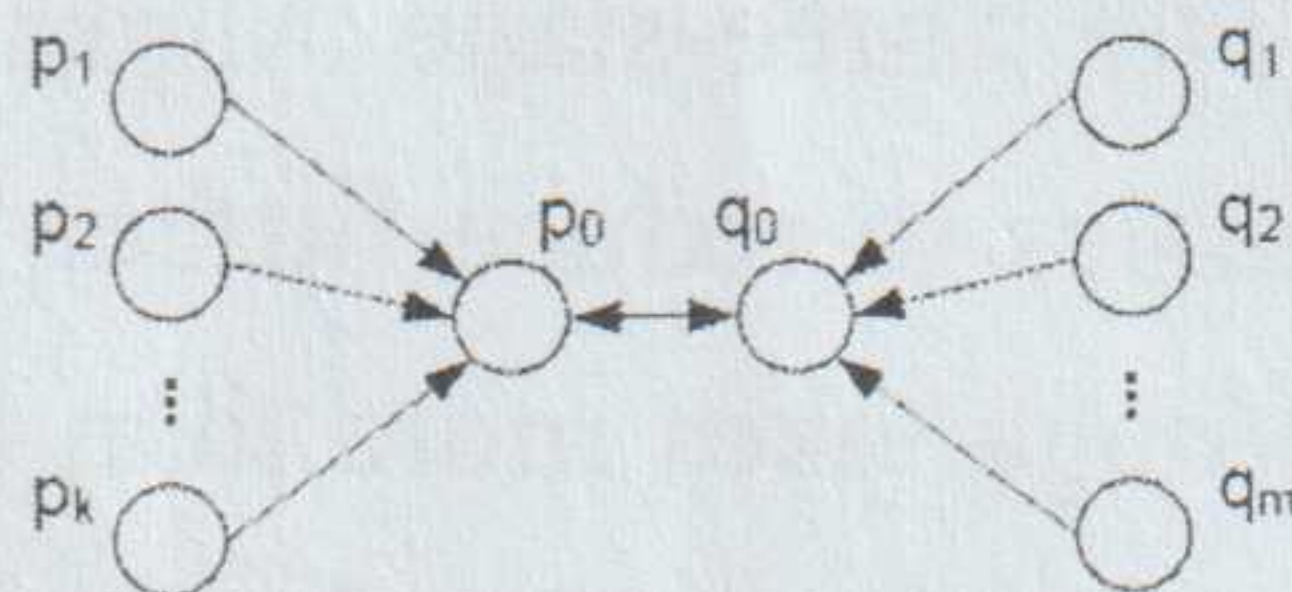
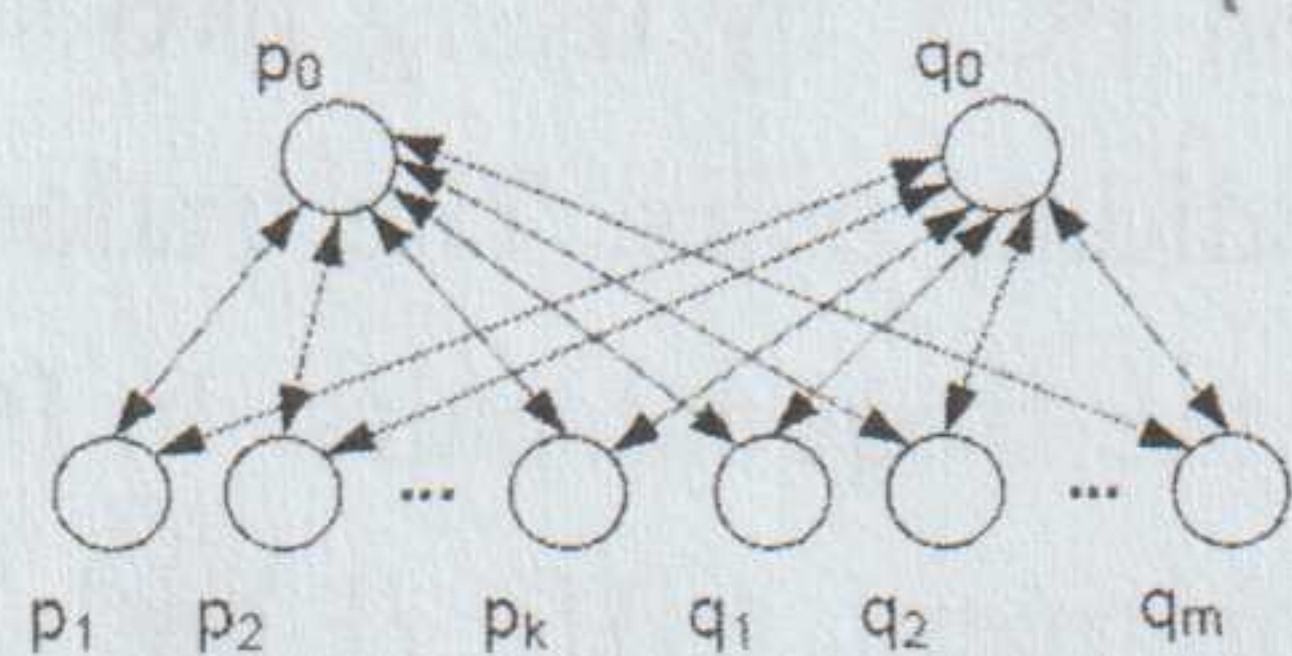
$N \dots \#$ pages on the web
 $M \dots \#$ of pages spammer owns

$$y = \sum \frac{\beta}{d} + \frac{1-\beta}{N}$$

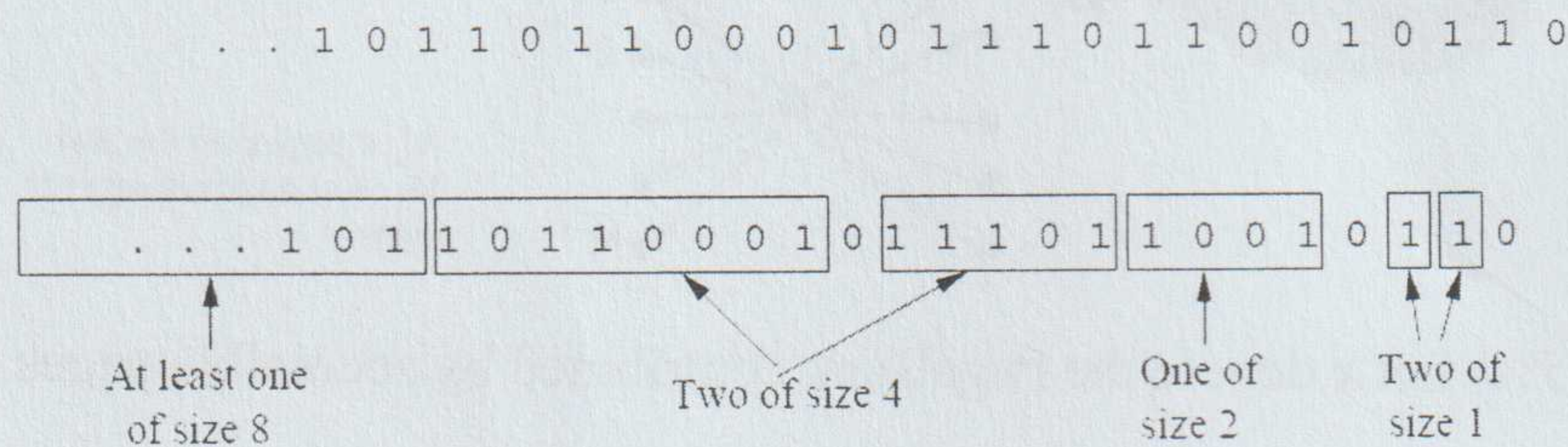
- (a) (5%) Let x denote the PageRank contributed by accessible pages and y be the PageRank of target page t . And rank of each "farm" page is $\frac{\beta y}{M} + \frac{1-\beta}{N}$. Show

that $y = \frac{\beta y}{1-\beta^2} + c \frac{M}{N}$, where $c = \frac{\beta}{1+\beta}$.

- (b) (3%) Suppose two spam farmers agree to link their spam farms. How would you link the pages in order to increase as much as possible the PageRank of each spam farm's target page? One way of doing it is just having all boosting pages point to both of the targets, as presented in left figure. Another way of connecting the two target pages and removing all links, as shown in the right figure. Which one performs better? Explain your reason with some calculations as stated by (a).



10. (10%) In the general case, the worst competitive ratio of the BALANCE (many bidders) on-line algorithm is $1 - 1/e \approx 0.63$.
- (a) (2%) Gives the worst case example with N advertisers of the same budget B , and NB queries in N rounds.
- (b) (2%) What's the optimum revenue in the off-line solution?
- (c) (6%) Show that the revenue of BALANCE is $BN(1 - 1/e)$, and thus the competitive ratio is $1 - 1/e$.
11. (8%) A demonstrated example of the simple DGIM algorithm (uses $O(\log^2 N)$ bits to represent a window of N bits with an error of no more than 50%.) is shown in the following figure. Consider $k = 10$. Then the query asks for the number of 1's in the ten rightmost bits, which happen to be 0110010110. Let the current timestamp (time of the rightmost bit) be t . Then the two buckets with one 1, having timestamps $t - 1$ and $t - 2$ are completely included in the answer. However, the rightmost bucket of size 4, with timestamp $t - 8$ is only partly included. DGIM estimate of the number of 1's in the last ten positions is thus 6. Of course the correct answer is 5.



- (a) (4%) Show that DGIM requires $O(\log^2 N)$ bits to represent a window of N bits.
- (b) (4%) The estimation error of the simple DGIM is no more than 50%, why?
12. (8%)

	MMDS	Machine Learning	Data Structures	Dynamics	Mechanics
Diane	1	1	1	0	0
Ethan	2	2	2	0	0
Frank	1	1	1	0	0
Grace	5	5	5	0	0
Hank	0	0	0	2	2
Ingrid	0	0	0	3	3
Joe	0	0	0	1	1

Matrix M represents the ratings of Engineering courses taken by Stanford students. Each row of M represents the given student's set of ratings and the columns of M represent the classes. The labels for students and classes are provided along each row and column. An entry M_{ij} in M represents the student i 's rating for class j . Now, the SVD decomposition of matrix M is found to be as follows:

$$USV^T = \begin{bmatrix} .18 & 0 \\ .36 & 0 \\ .18 & 0 \\ .90 & 0 \\ 0 & .53 \\ 0 & .80 \\ 0 & .27 \end{bmatrix} \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \begin{bmatrix} .58 & .58 & .58 & 0 & 0 \\ 0 & 0 & 0 & .71 & .71 \end{bmatrix}$$

It's pretty clear that there are two concepts of classes here: the first three are Computer Science classes while the last two are Mechanical Engineering classes.

Suppose a new student named Tony has the following reviews: 4 for Machine Learning, 5 for Data Structures, and 2 for Dynamics. This gives a representation of Tony in the class space as $[0 \ 4 \ 5 \ 2 \ 0]$.

- (a) (4%) What is the representation of Tony in the concept space?
 - (b) (4%) Explain what this representation predicts about how much Tony would like the two remaining classes that he has not reviewed (MMDS and Mechanics).
13. (5%) Suppose we are given a matrix R of recommendations. The element R_{iu} of this matrix corresponds to the rating given by user u to item i . The size of R is $m \times n$, where m is the number of movies, and n the numbers of users. Most of the elements of the matrix are unknown because each user can only rate a few movies. Our goal is to find two matrices P and Q , such that $R \approx QP^T$. The dimensions of Q are $m \times k$, and the dimensions of P are $n \times k$. k is a parameter of the algorithm. We define the error as

$$E = \sum_{(i,u) \in \text{ratings}} (R_{iu} - q_i \cdot p_u^T)^2 + \lambda \left[\sum_u \|p_u\|_2^2 + \sum_i \|q_i\|_2^2 \right].$$

Let ϵ_{iu} denote the derivative of the error E with respect to R_{iu} . What is the expression for ϵ_{iu} ? What are the update equations for q_i and p_u in the Stochastic Gradient Decent algorithm?

14. (4%) Suppose we wish to store an $n \times n$ boolean matrix (0 and 1 elements only). We could represent it by the bits themselves, or we could represent the matrix by listing the positions of the 1's as pairs of integers, each integer requiring $\lceil \log_2 n \rceil$

Handwritten notes:
 $1 \leq i \leq n$
 $1 \leq j \leq n$
 $n \times n$
 $n \times n \times \log n$
 (i, j)
 $n \times n \times \log n$

bits. The former is suitable for dense matrices; the latter is suitable for sparse matrices. How sparse must the matrix be (i.e., what fraction of the elements should be 1's) for the sparse representation to save space?

15. (6%) How could we use a construction on the family of minhash functions to transform a $(0.2, 0.8, 0.8, 0.2)$ -sensitive family into a $(0.2, 0.8, 0.9991285, 0.0000004)$ -sensitive family?

(a) (4%) Describe such a construction.

(b) (1%) What's the number of minhash functions in this case.

(c) (1%) The resulting fault positive rate is ____.

$$\begin{aligned} (d_1, d_2) \leq 0.2 &\rightarrow p \geq 0.8 \\ (d_1, d_2) \geq 0.8 &\rightarrow p \leq 0.2 \\ &\rightarrow p \geq (1 - 0.8^n) \end{aligned}$$

16. (6%) For cosine distance, one can use the Random Hyperplanes method to construct $(d_1, d_2, (1 - d_1/\pi), (1 - d_2/\pi))$ -sensitive LS-family for any d_1 and d_2 . In this method, this LS-family is set of all functions derived from any vector, and each vector $\mathbf{v}$ determines a hash function $h_{\mathbf{v}}$ with two buckets, $h_{\mathbf{v}}(\mathbf{x}) = +1$ if $\mathbf{v} \cdot \mathbf{x} \geq 0$; $h_{\mathbf{v}}(\mathbf{x}) = -1$ if $\mathbf{v} \cdot \mathbf{x} < 0$. Show that for points $\mathbf{x}$ and $\mathbf{y}$, $\Pr[h(\mathbf{x}) = h(\mathbf{y})] = 1 - d(\mathbf{x}, \mathbf{y})/\pi$.

17. (8%) The k -armed bandit problem as a formalization of the exploration-exploitation tradeoff. Some simple algorithms are able to achieve no regret, such as Epsilon-greedy and Upper Confidence Sampling (UCB). Describe the UCB algorithm, where the upper confidence interval (based on the Hoeffding's inequality) is denoted as

$$UCB(a) = \hat{\mu}_a + \alpha \sqrt{\frac{2 \ln t}{m_a}}$$

$$\begin{aligned} 0.04 \\ 0.08 \\ 0.0016 \\ (2 \times 10^{-4})^r &\leq 4 \times 10^{-7} \\ \rightarrow 2^r \times 10^{-r} &\leq 2^2 \times 10^{-7} \\ r(\log 2 - 1) &\leq \log 4 - 7 \\ r &\leq \frac{\log 4 - 7}{\log 2 - 1} = \frac{0.6}{-0.7} \\ &= -0.857 \end{aligned}$$

$$\begin{aligned} 5.10 \\ 0.6 \\ 6.4 \end{aligned}$$

國立清華大學試卷

記		分	
1	16	2	1
3	4	4	2+2
5	0	6	0
7	2	8	1
9	7	10	4
11	5	12	8
13	4	14	4
15	0	16	6
17	0	18	
19		20	
總分		64+2=66	

所系 EE

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日期 12/9

1. (a) X (d) ~~0~~ (g) 0 (j) X
 (b) X (e) X (h) X (k) ~~X~~
 (c) 0 (f) ~~0~~ (i) 0 (l) ~~X~~ -8

2. ~~frequent itemsets~~, ~~similar items~~, clustering web mining -3

3. Let $y = (1 + \frac{1}{x})^x \Rightarrow \ln y = x \ln(1 + \frac{1}{x}) = \frac{\ln(1 + \frac{1}{x})}{\frac{1}{x}}$

note that y is a function of x ,

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{1}{x})}{\frac{1}{x}} \xrightarrow{\text{L'Hospital rule}} \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{x}} \cdot (-\frac{1}{x^2})}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1$$
+4

$\Rightarrow$ When $x \rightarrow \infty$, $y = (1 + \frac{1}{x})^x \rightarrow e = e$ ✕

4.

	C ₁	C ₂	C ₃	C ₄
R ₃	0	1	1	0
R ₄	0	1	1	0
R ₇	1	0	0	1
R ₂	1	0	1	0
R ₆	1	0	0	1
R ₁	1	0	0	1
R ₅	0	1	1	0

→ minhash

C ₁	C ₂	C ₃	C ₄
1	3	3	1

$$\text{similarity} = \frac{|C_i \cap C_j|}{|C_i \cup C_j|} = \text{SIM}(C_i, C_j)$$

(i, j)	col/col SIM	sig/sig SIM	(i, j)	col/col SIM	sig
1 1	1	1	3 3	1	
1 2	0	0	3 4	0	
1 3	1/7	0	4 4	1	
1 4	3/4	1			
2 2	1	1			
2 3	3/4	1			
2 4	1/0	0			

+2+2

$$6. \text{conf}(S \rightarrow T) = \frac{\text{support}(S \cap T)}{\text{support}(S \cup T)}$$

$\therefore$ No greater than $\text{conf}(\{1,2,3,4\} \rightarrow \{5\})$

$\therefore$ (a) 不行

$\therefore$ as great as $\text{conf}(\{1,2\} \rightarrow \{3,4,5,6,7\})$

$\therefore$ (c) 可以 (d) 不行

$\Rightarrow$ Ans: (a) (d) ~~(b)~~ +0

7. 若通过 (cc), 则 (i,j) 只要 hash 到 basket 2 就被 count, 但是 i, j 各为 frequent item, 并不保证 (i,j) 必为 frequent pair +2

8. ① ~~b~~ f ④ ~~e~~ ⑦ ~~d~~ a
 ② ~~a~~ b ⑤ ~~e~~ c
 ③ g ⑥ ~~e~~ d -b

9. (a) By page rank equation,

$$y = x + \beta M \left(\frac{\beta y}{N} + \frac{1-\beta}{N} \right) + \frac{1-\beta}{N}$$

1 sig SIM
 1
 0
 1

$$\Rightarrow y = x + \beta^2 y + \frac{M}{N} \beta (1-\beta)$$

$$\Rightarrow (1-\beta^2) y = x + \frac{M}{N} \beta (1-\beta)$$

$$\Rightarrow y = \frac{x}{1-\beta^2} + \frac{\beta}{1+\beta} \frac{M}{N}$$

relative small ignore.

(b) 左圖中, page rank 值若為 r_{p_0}, r_{g_0} , take $\beta = 0.8$

$$r_{p_0} = \sum_{i \rightarrow p_0} \beta \frac{r_i}{d_i} + \frac{1-\beta}{N}$$

$$= \frac{\beta}{2} \left(\sum_{i=1}^k r_{p_i} + \sum_{i=1}^m r_{g_i} \right) + \frac{1-\beta}{N}$$

令 $\sum_{i=1}^k r_{p_i} = P, \sum_{i=1}^m r_{g_i} = Q$, 則

$$r_{p_0} = \frac{2}{5}(P+Q) + \frac{1}{5} \frac{1}{N} = r_{g_0}$$

算式同 p_0

右圖中, 依同理計算

$\triangleq P'$

$$r_{p_0} = \sum_{i \rightarrow p_0} \beta \frac{r_i}{d_i} + \frac{1-\beta}{N} = \beta \left[\sum_{i=1}^k r_i \right] + \beta r_{g_0} + \frac{1-\beta}{N}$$

$$r_{g_0} = \sum_{i \rightarrow g_0} \beta \frac{r_i}{d_i} + \frac{1-\beta}{N} = \beta \left[\sum_{i=1}^m r_i \right] + r_{p_0} + \frac{1-\beta}{N}$$

$$\Rightarrow r_{p_0} = \beta P' + \beta (\beta Q' + r_{p_0} + \frac{\beta(1-\beta)}{N}) + \frac{1-\beta}{N}$$

$$\Rightarrow (1-\beta)r_{p_0} = \beta P' + \beta^2 Q' + \frac{\beta^2(1-\beta)}{N}$$

$$\Rightarrow r_{p_0} = \frac{\beta}{1-\beta} P' + \frac{\beta^2}{1-\beta} Q' + \frac{\beta^2}{N} = 4P' + 3.2Q' + \frac{0.64}{N}$$

同理, $r_{g_0} = 4Q' + 3.2P' + \frac{0.64}{N}$

故若 $P' \approx Q' \approx P \approx Q$ 時, 以左圖為佳

10. (a) Suppose there are N advertisers, say $A_1, \dots, A_N$, and NB queries, in each query q_i , q_i has $N-i+1$ advertisers $A_i, A_{i+1}, \dots, A_N$

(b) NB

(c) In the worst case, the revenue is

$$\begin{aligned}
 & NB - B \left(\frac{1}{N} + \frac{1}{N-1} + \dots + \frac{1}{N-i+1} \right) \\
 &= NB \left(1 - \left(\frac{1}{N} + \frac{1}{N-1} + \dots + \frac{1}{N-i+1} \right) \right) \\
 &\geq NB \left(1 - \frac{1}{e} \right)
 \end{aligned}$$

Euler's approximation

$$\Rightarrow \text{competitive ratio} = \frac{BN(1-\frac{1}{e})}{BN} = 1 - \frac{1}{e}$$

11 (a) If there is N bits, 則需要 $O(\log N)$ bucket, 又每個 bucket 需儲存 " n " 這個數字, n 為 "1" 的数目, 若 n 以二進位儲存,

$$\Rightarrow O(2^n) = O(N) \Rightarrow O(n) = O(\log N) \quad +1$$

$\Rightarrow$ 需 $O(\log N)$ bit 來儲存 n

for any bucket,

$$\Rightarrow \text{共需 } O(\log N) \times O(\log N) = O(\log^2 N) \text{ bit}$$

(b).

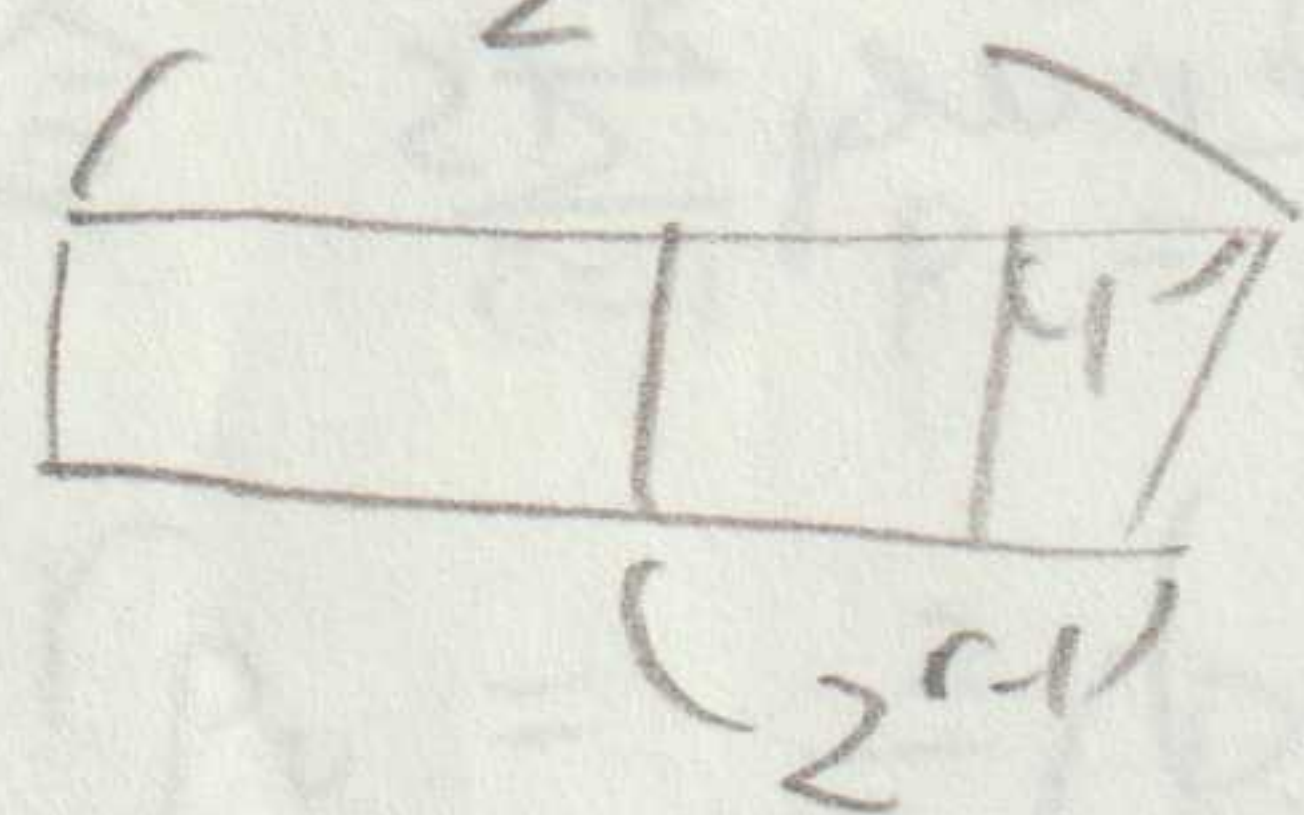
data stream

bucket of size 2^r bucket of size 2^{r-1}

如前頁圖所示, 假設最左端之 bucket 為 size 2^r ,

則求 $\max \text{ error} = \frac{\max(\text{size } 2^r \text{ bucket 造成之 error})}{\min(\text{已算過之 bit 數, 含 size } 2^r \text{ 之 bucket})}$

$$= \frac{2^r - 2^{r-1} - 1}{1 + 2 + \dots + 2^{r-1} + 1} = \frac{2^{r-1}(2-1) - 1}{(2^r - 1) + 1}$$



size 2^r 之
bucket 至少
含 1 bit

$$= \frac{2^{r-1} - 1}{2^r}$$

+4

$$= \frac{1}{2} - \frac{1}{2^r} < \frac{1}{2} \quad \times$$

12.(a)

所求即

$$[0.4 \ 5.2 \ 0]$$

$$\begin{bmatrix} 0.58 & 0 \\ 0.58 & 0 \\ 0.58 & 0 \\ 0 & 0.71 \\ 0 & 0.71 \end{bmatrix}$$

2.32

2.90

1.42

$$= [5.22 \ 1.42]$$

+8

(b) Tony would prefer MADS rather than Mechanics since 5.22 (preference of CS class) > 1.42 (preference of Mechanics) $\times$

13. W.L.O.G, $\because \lambda$ 為 hyper parameter, 假設以 $\frac{\lambda}{2}$ 取代 λ , 則

$$E = \sum_{(i,u)} (R_{iu} - g_i p_u^T)^2 + \frac{\lambda}{2} \left(\sum_u \|p_u\|_2^2 + \sum_i \|g_i\|_2^2 \right)$$

$$\Rightarrow E_{iu} = \frac{\partial E}{\partial R_{iu}} = \sum_{(i,u)} 2(R_{iu} - g_i p_u^T)$$

To determine the update equation, calculate the gradients

$$\begin{aligned}\nabla f_i(R_{iu}) &= -2(R_{iu} - g_i \cdot \mathbf{p}_i^T) \mathbf{p}_i^T + 2g_i \\ &= -(\epsilon_{iu} \mathbf{p}_i^T - 2g_i)\end{aligned}$$

$$\begin{aligned}\nabla p_u(R_{iu}) &= -2(R_{iu} - g_i \cdot \mathbf{p}_i^T) g_i + 2p_u^T \\ &= -(\epsilon_{iu} g_i - 2p_u^T)\end{aligned}$$

A SGD update equation is

$$\begin{cases} g_i \leftarrow g_i - \mu_1 \nabla f_i(R_{iu}) \\ \mathbf{p}_i^T \leftarrow \mathbf{p}_i^T - \mu_2 \nabla p_u(R_{iu}) \end{cases}$$

$$\Rightarrow \begin{cases} g_i \leftarrow g_i + \mu_1 (\epsilon_{iu} \mathbf{p}_i^T - 2g_i) \\ \mathbf{p}_i^T \leftarrow \mathbf{p}_i^T + \mu_2 (\epsilon_{iu} g_i - 2p_u^T) \end{cases}$$

14. In the "dense" matrix, we need $n \times n$ bits in the memory

In the "sparse" matrix, suppose the fraction of 1's is p , we need $n \times n \times p \times \lceil \log n \rceil \times 2$ bits in the matrix

save space $\Rightarrow n \times n \times p \times \lceil \log n \rceil \times 2 < n \times n$

$$\Rightarrow p < \frac{1}{\lceil \log n \rceil \times 2}$$

15. 令 $(0.2, 0.8, 0.8, 0.2)$ -sensitive 2 minhash function

family 为 $H = \{h_1, h_2, \dots, h_n\}$ 定义一个新的 minhash function family H' , 其中的元素 h 为

$(h_{i_1}, h_{i_2}, \dots, h_{i_r}) \subset H$, 且每个 hashing function

独立工作

其中 r 是使 $1 - 0.8^r \geq 0.9991285$ 之最大整数

$$\{0.2^r \leq 0.0000004\}$$

(b) $\because 0.2^r \leq 0.0000004$

$$\therefore \text{取 } r \log(2 \times 10^{-1}) = \log(4 \times 10^{-7})$$

$$\Rightarrow r = \left\lceil \frac{2 \log 2 - 7}{\log 2 - 1} \right\rceil + 1, \text{ 取 } \log 2 \doteq 0.3$$

$$\Rightarrow r = 10$$

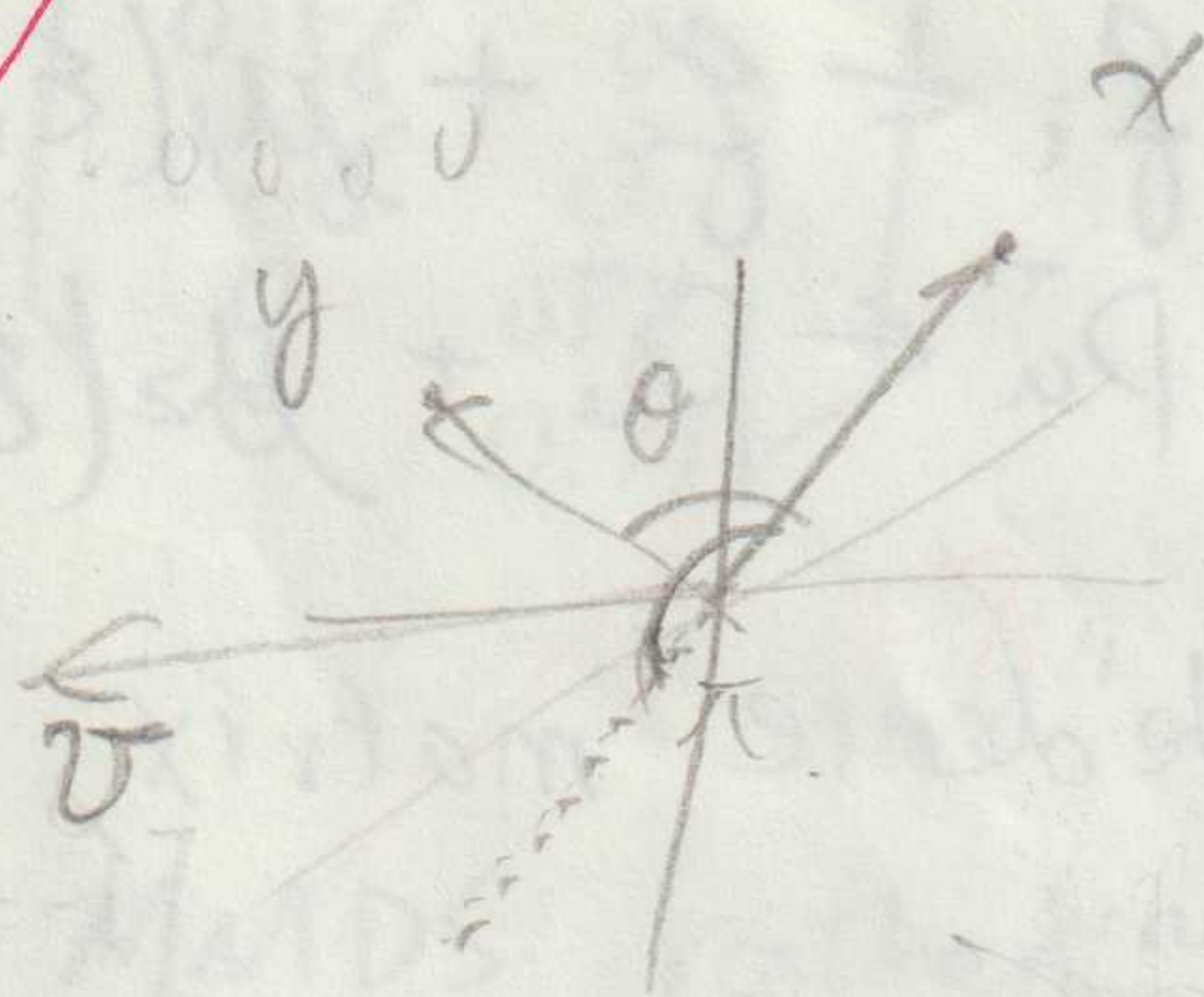
(c) $1 - 0.9991285 = 0.0008715$

16. $h_v(x) = \begin{cases} 1, & v \cdot x \geq 0 \\ -1, & v \cdot x < 0 \end{cases}$

$$P[h(x) = h(y)]$$

$$= 1 - P[h(x) \neq h(y)]$$

$$= 1 - \frac{d(x, y)}{\pi}$$



17 不会 QQ 5 也不会, 呜呜

-8